



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Computer & Systems Sciences
Name of Programme / Course:	OGE
Semester/Year:	Semester-IV
Paper Code:	SCSSAGE401T22
Title of Paper:	End User Database Management System
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections. Section A and Section B.
2. Section A consists of seven questions and you have to attempt any five questions.
3. Section B consists of six questions and you have to attempt any four questions.
4. Marks allotted are mentioned against each question.
5. Draw neat and clean diagrams wherever required.

Question Number	SECTION-A	Marks Allotted
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- Q.1 What are the drawbacks of the **file-based system** when compared to a **Database Management System (DBMS)**? Discuss at least three disadvantages. 5X4=20
- Q.2 List and briefly explain any **four characteristics** of the database approach in comparison to traditional file-based systems.
- Q.3 What is the difference between **data** and **information**? Provide examples to illustrate the distinction.
- Q.4 What are **entity types** and **relationships** in the context of **Entity-Relationship (ER) Modeling**? Provide an example of an ER diagram involving one entity and one relationship.
- Q.5 What is **SQL schema definition**? Write a simple SQL statement to define a table for storing employee details (employee_id, name, and salary).

- Q.6 Define **Primary Key** and **Foreign Key** in the context of the relational data model. How do these keys ensure relational integrity?
- Q.7 Explain First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). Provide an example of a table that satisfies 1NF but not 2NF.

Section-B

- Q.1 Explain the **DBMS architecture** and how **data independence** 4X10=40 is achieved in a DBMS. Describe the difference between **logical data independence** and **physical data independence** with examples.
- Q.2 Discuss the **Entity-Relationship (ER) model** and explain how **Enhanced ER modeling** improves the basic ER model. Provide an example of an ER diagram with at least two entities, their relationships, and attributes.
- Q.3 What are the **ACID properties** in database systems? Explain how each of the ACID properties (Atomicity, Consistency, Isolation, Durability) ensures database reliability and integrity.
- Q.4 What are **relational integrity** constraints? Explain **Codd's 12 rules** for relational databases. Pick any three of Codd's rules and explain them in detail with examples.
- Q.5 Discuss the process of **normalization** in database design. Explain **Fourth Normal Form (4NF)** and **Fifth Normal Form (5NF)** with examples. How do these higher normal forms help in reducing redundancy?
- Q.6 Describe the three levels of database design: **Conceptual Design**, **Logical Design**, and **Physical Design**. Discuss the significance of each level and how they contribute to creating an efficient database system.